

|                                                    |                                             |
|----------------------------------------------------|---------------------------------------------|
| Program : <b>Diploma in Biomedical Engineering</b> |                                             |
| Course Code : <b>6242A</b>                         | Course Title: <b>Healthcare Informatics</b> |
| Semester : <b>6</b>                                | Credits: <b>4</b>                           |
| Course Category : <b>Open Elective</b>             |                                             |
| Periods per week: <b>4 (L:4 T:0 P:0)</b>           | Periods per semester : <b>60</b>            |

### Course Objectives:

- To introduces the requirements of a healthcare information systems.
- To provide the basic skills and knowledge of informatics in healthcare field.

### Course Prerequisites:

| Topic                          | Course code | Course Title                      | Semester |
|--------------------------------|-------------|-----------------------------------|----------|
| Basic concepts about computers |             | Introduction to IT systems<br>Lab | 1        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                                         | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Summarize the structure of a hospital and outline hardware system requirements for hospital information management. | 16               | Understanding   |
| CO2             | Illustrate the architecture of IT systems in hospitals and outline the regulations related to health information.   | 13               | Understanding   |
| CO3             | Exemplify medical database, electronic patient record, data collection, analysis methods.                           | 15               | Understanding   |
| CO4             | Describe the pharmacy information system, clinical laboratory informatics and radiology information system          | 14               | Understanding   |
|                 | Series Test                                                                                                         | 2                |                 |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 2   |     |     |     |     |     |     |
| CO3             | 2   |     |     |     |     |     |     |
| CO4             | 2   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                                                                                                                | Duration (Hours) | Cognitive Level |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Summarize the structure of a hospital and outline hardware system requirements for hospital information management.</b> |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Summarize the structure of a hospital                                                                                      | 5                | Understanding   |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Define health informatics, medical informatics, bioinformatics, data, information and knowledge                            | 6                | Remembering     |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Identify the components and outline hardware requirements and networking in hospitals                                      | 5                | Understanding   |
| <b>Contents:</b><br>General Organization of a hospital, departments in a hospital – clinical and nonclinical , process flow<br>Definitions of health informatics, medical informatics, bioinformatics, data, information and knowledge, converting data into knowledge<br>System components of health informatics-hardware requirements in a hospital, types of networking in hospitals, Clinical data warehouse, challenges in healthcare informatics |                                                                                                                            |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Illustrate the architecture of IT systems in hospitals and outline the regulations related to health information.</b>   |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Illustrate the information technology architecture models in healthcare organizations                                      | 5                | Understanding   |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Summarize the healthcare information regulations and standards                                                             | 5                | Understanding   |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Describe the classification and coding of medical system                                                                   | 3                | Understanding   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Series Test I                                                                                                              | 1                |                 |
| <b>Contents:</b><br>Information technology architecture models in healthcare organization, basic architecture of hospital information system.                                                                                                                                                                                                                                                                                                          |                                                                                                                            |                  |                 |

Legal aspects of managing health information, health records and privacy, health level standard - HL7

Classification and coding of medical systems - principles of classification and types of classifications.

|            |                                                                                                  |   |               |
|------------|--------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Exemplify medical database, electronic patient record, data collection, analysis methods.</b> |   |               |
| M3.01      | Describe the medical databases : EMR and EHR                                                     | 4 | Understanding |
| M3.02      | Illustrate the functional components of EHR                                                      | 4 | Understanding |
| M3.03      | Exemplify the process of data collection, data analysis                                          | 7 | Understanding |

**Contents:**

Computerized Patient Records

Introduction to medical databases: Electronic Medical Record, Electronic health record, Electronic health record vs paper based record,

Functional Components of an Electronic Health Record System

Data collection, data analysis, health information exchange, interfaces, challenges in EHR

Case study: Create a sample EHR for an individual.

|            |                                                                                                                    |   |               |
|------------|--------------------------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO4</b> | <b>Describe the pharmacy information system, clinical laboratory informatics and radiology information system.</b> |   |               |
| M4.01      | Describe the structure of hospital information system                                                              | 4 | Understanding |
| M4.02      | Summarize the pharmacy information system and laboratory information system                                        | 6 | Understanding |
| M4.03      | Explain the radiology information system and PACS                                                                  | 4 | Understanding |
|            | Series Test II                                                                                                     | 1 |               |

**Contents:**

Hospital Information System

Hospital Information System- Definition, Structure and application , advantages

Automated clinical laboratory information system , Pharmacy information system

Radiology information system, Picture Archival and communication system architecture, protocols

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                                                                               |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1         | Hoyt, R. E., Yoshihashi, A. K. (2014). Health informatics: Practical guide for healthcare and information technology professionals (6th ed.). Informatics Education.                   |
| T2         | Shortliffe, Edward H., et al.(2014) Biomedical informatics: computer applications in health care and biomedicine(4 <sup>th</sup> ed). Springer.                                        |
| R1         | Wager, K. A., Lee, F. W., & Glaser, J. P. (2017). Health care information systems: A practical approach for health care management -4th Edition. Jossey-Bass. ISBN-13: 978-1119337188. |
| R2         | S A Kelkar(2010) Hospital Information systems- A concise study , Prentice Hall of India                                                                                                |
| R3         | Braunstein, M. L. (2014). Contemporary Health Informatics. American Health Information Management Association.                                                                         |

**Online resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://en.wikipedia.org/wiki/Health_information_management">https://en.wikipedia.org/wiki/Health_information_management</a>                             |
| 2            | <a href="https://www.himss.org/resources/healthcare-informatics">https://www.himss.org/resources/healthcare-informatics</a>                                       |
| 3            | <a href="http://nhsrindia.org/health-informatics#">http://nhsrindia.org/health-informatics#</a>                                                                   |
| 4            | <a href="https://drdollah.com/radiology-information-system-and-pacs/">https://drdollah.com/radiology-information-system-and-pacs/</a>                             |
| 5            | <a href="https://drdollah.com/hospital-information-system-his/system-architecture/">https://drdollah.com/hospital-information-system-his/system-architecture/</a> |